EWDA 2026 Conference – Novi Sad, Serbia, 29.IX.–03.X.2026

Post-release monitoring in rehabilitated wildlife: Insights from a multi-species telemetry pilot study in Germany

Natalie Steiner[1], Florian Brandes[3], Karolin Schütte[3], Lena Rief[2], Ulrich Voigt[1], Daniel Knitter[4], Jonas Höchst[4], Tobias Petschinka[4], Jannis Gottwald[4], Ursula Siebert[1,5,6] and Michael Pees[2]

[1] Institute for Terrestrial and Aquatic Wildlife Research (ITAW), University of Veterinary Medicine Hannover, Germany
[2] Department of Small Mammal, Reptile and Avian Diseases, University of Veterinary Medicine Hannover, Germany
[3] Wildlife Rescue and Conservation Center, Sachsenhagen, Germany
[4] trackIT Systems GmbH, Cölbe, Germany
[5] Leibniz Institute for Zoo and Wildlife Research, Berlin, Germany
[6] Department of Ecoscience, Aarhus University, Aarhus, Denmark

BACKGROUND

- Wildlife rehabilitation is costly and widespread – but post-release survival and behaviour remain largely unknown
- Without outcome data, protocols cannot be improved or evaluated

AIM

Test whether telemetry can reliably track post-release survival and behavioural reintegration in multiple rehabilitated German wildlife species.

MATERIAL AND METHODS

STUDY DESIGN – SPECIES AND TAG TYPES

98 individuals tagged over two years (2024–2026), across **five species** groups.

Common buzzard
Buteo buteo
n = 14 **GPS**

Edible dormice
Glis glis
n = 20 **VHF + RFID**

European hedgehog
Erinaceus europaeus
n = 26 **VHF**

House sparrow
Passer domesticus
n = 19 **VHF**

Thrush species
Turdus spp.
n = 19 **VHF**

STUDY SITE AND AUTOMATED ARRAY SYSTEM

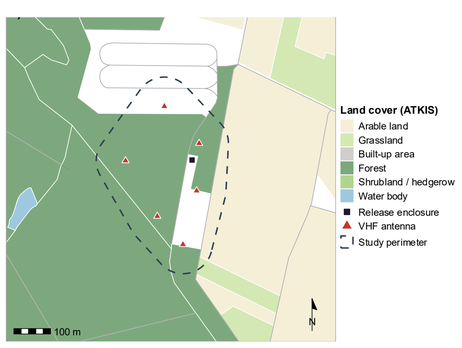


Fig. 1 Study site with setup of automated VHF array.

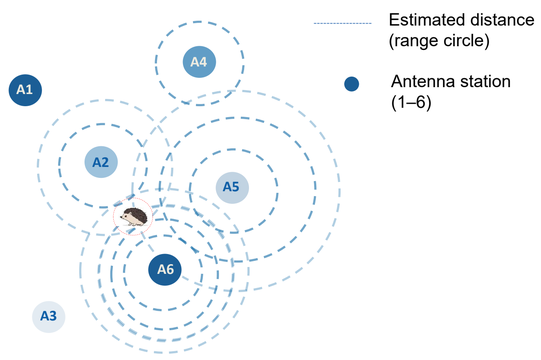
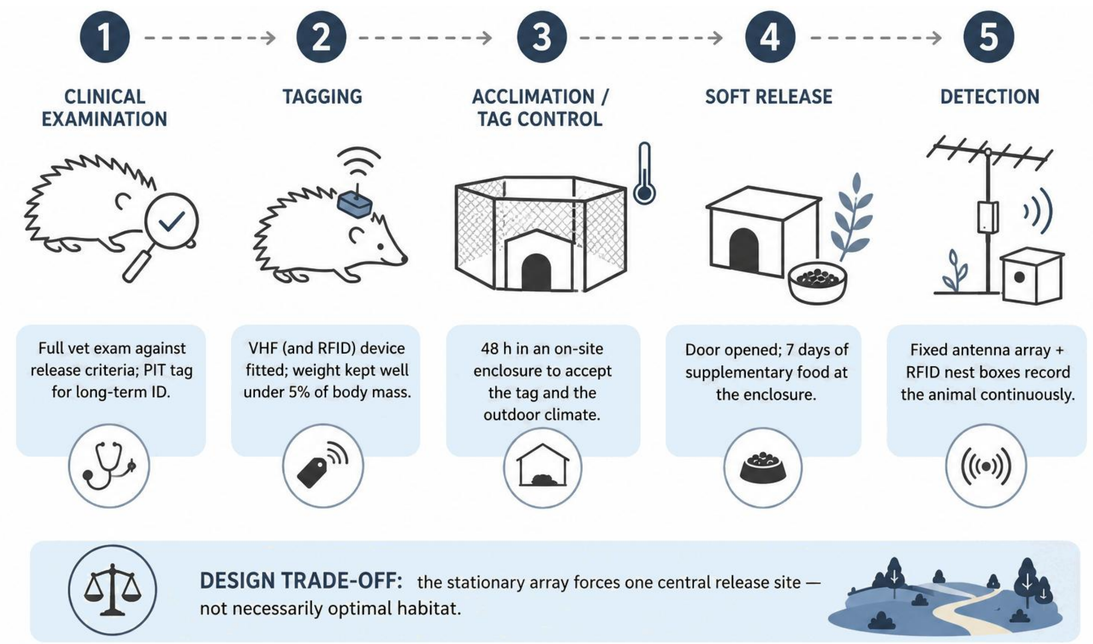


Fig. 2 Illustration of the antenna array system using multilateration.

DATA ANALYSIS random forest & cosinor | aKDE home ranges | Jacobs' D electivity | Minimum convex polygon (MCP) | Kaplan-Meier & Cox regression

RELEASE PROTOCOL

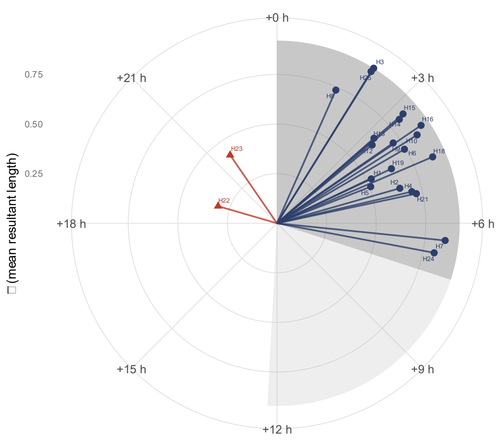


FIRST RESULTS – HEDGEHOGS & BUZZARDS

European hedgehog – *Erinaceus europaeus*

Circadian activity returns from the first night
Within-night structure reorganizes
- Number of activity bouts decreased significantly (GLMM)
- While mean bout duration stayed constant

Site fidelity: Animals stayed close to release site (approx. 150 m)
Space use: Edge habitat was strongly selected
Visual impairment significantly disrupts rhythmicity (Figs. 3 and 4)



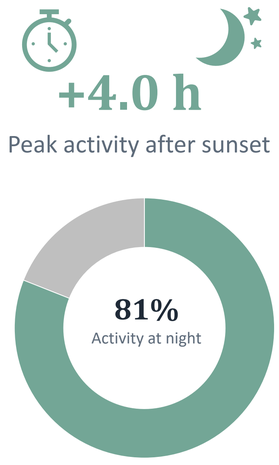


Fig. 3 Rayleigh plot of mean activity time relative to sunset; arrows = individual mean vectors, colour = visual status.

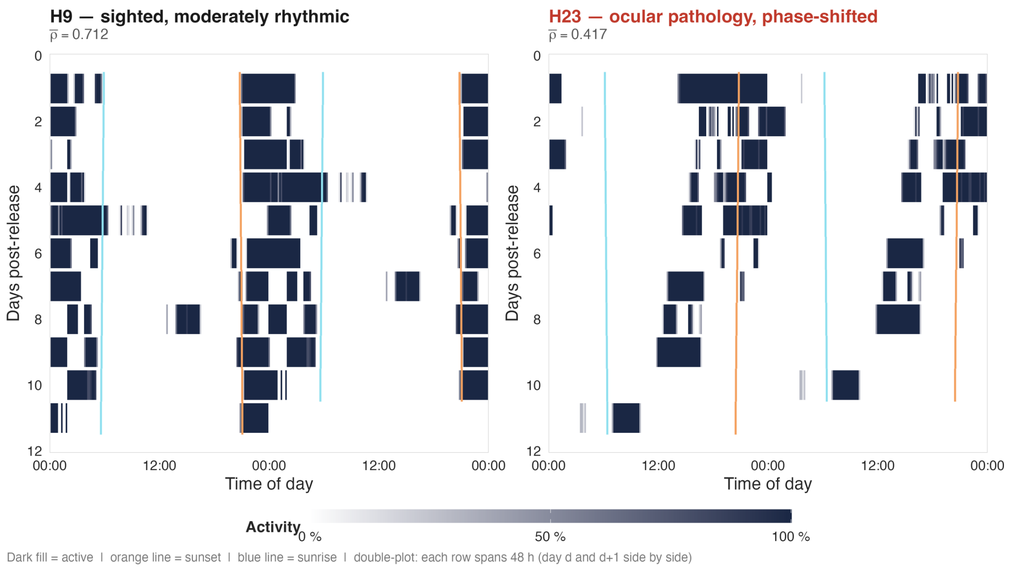
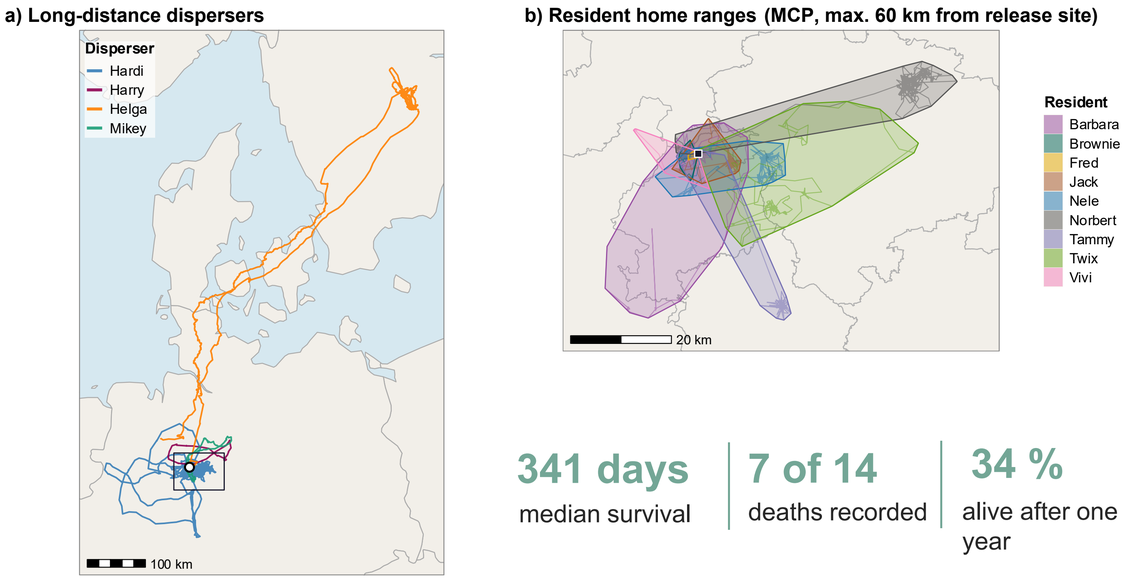


Fig. 4 Double-plotted actograms of a sighted (H9) and a visually impaired (H23) hedgehog over days post-release; darker = higher activity.

Common buzzards – *Buteo buteo*



341 days median survival | **7 of 14** deaths recorded | **34 %** alive after one year

Fig. 5 Post-release movements of 13 GPS-tagged common buzzards. (a) Four long-distance dispersers; black box = area shown in (b). (b) Nine residents that remained within 60 km of the release site, with minimum convex polygon (MCP) home ranges; polygons include commuting flights between the release site and the area of origin. Tracks censored at the last signal from the living bird. Data as of 13 Sept 2026.

CONCLUSION & OUTLOOK

High-resolution automated VHF telemetry reveals fine-scale behaviour with implications for wildlife rehabilitation.

Rehabilitation success should be defined behaviourally, integrated with welfare – One Welfare approach.

Species-specific protocols, pre-release screening, refined telemetry attachment and release mechanisms, as well as networking needed.

REFERENCES

Mullineaux, E., 2014. Veterinary treatment and rehabilitation of indigenous wildlife. J. Small Anim. Pract. 55, 293–300. https://doi.org/10.1111/jsap.12213.
Burroughes, N.D., Dowler, J., Burroughes, G., 2021. Admission and survival trends in hedgehogs admitted to RSPCA wildlife rehabilitation centres. Proc. Zool. Soc. 74, 198–204. https://doi.org/10.1007/s12595-021-00363-9.

CONTACT natalie.steiner@tiho-hannover.de
natalie-steiner@hotmail.com

EWDA 2026 CONFERENCE NOVI SAD, SERBIA

